MILITARY SPECIFICATION

OBTURATING BAND, PLASTIC SPLIT (MOLDED)

This Specification is approved for use within the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements and quality assurance provisions for obturating band, plastic, split (molded) for use on the Cartridge, 81mm; Illuminating, M853; Smoke, RP, M819 and Practice, M879.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition Standard Quality Assurance Provisions, General Specification for

STANDARDS

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U S Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:42Z
Check the source to verify that this is the current version before use.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings, and publications. The following other government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.3)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9310339 - Ring, Obturating

(Copies of specifications, standards, handbooks, drawings, and publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the document not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF TESTING AND MATERIAL (ASTM)

ASTM-D256	- Tests for Impact Resistance of Plastics and Electrical Insulating Materials
ASTM-D638	- Test for Tensile Properties of Plastics
ASTM-D1238	- Test for Flow Rates of Thermoplastics by Extrusion Plastometer
ASTM-D790	- Tests for Flexural Properties of Plastics and Electrical Insulating Materials
ASTM-D648	- Test for Deflection Temperature of Plastics Under Flexural Load
ASTM-D3935-80	- Polycarbonate (PC) Unfilled and Reinforced Materials

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:42Z
Check the source to verify that this is the current version before use.

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with applicable drawings, specifications, and this specification.

3.1.1 Certification. Certification must be furnished by the molder for each lot of bands certifying that the plastic used in molding complies with the requirements of 3.1.

3.2 Molded parts. Obturators shall be made in accordance with the applicable drawings and shall be produced by the injection molding process.

3.3 Color and transparency. Molded parts shall have green color as specified in accordance with the drawing and shall be sufficiently translucent to inspect for absence of voids.

3.4 Physical property of molded parts. Obturators shall meet the properties listed as follows (ASTM-D3935-80):

Characteristic	Requirements
Izod impact strength (min)	750 Joules/meter (J/M)
Tensile yield strength (min)	60 Megapascal (MPa)
Flow rate	4.5 gram/10 minute
Flexural Modulus (min)	2200 Megapascal (MPa)
Deflection temperature under a load at 1830 kilopascal (KPa)	130 degrees centigrade

3.5 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.6 Workmanship. The parts shall be uniform in color, free of cracks, voids, burn marks, flashing, contamination, and sprues.

4. QUALITY ASSURANCE PROVISIONS

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:42Z
Check the source to verify that this is the current version before use.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The following types of inspection shall be conducted on this item:

a. First article inspection.

b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the items as required per Table I and shall equally represent all mold cavities.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:42Z
Check the source to verify that this is the current version before use.

TABLE 1. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-O-70548 (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	Ring, Obturating	1 OF 1	See Below
			NEXT HIGHER ASSEMBLY: N/A

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Ring, Obturating (Dwg. 9310339) Examination for Defects	100		3.2/3.3	4.4.2.1
	Physical Properties of Molded Parts				
	Izod impact strength (min)	20		3.4	4.4.5.1.1
	Tensile yield strength	20		3.4	4.4.5.1.2
	Flow rate	20		3.4	4.4.5.1.3
	Flexural Modulus	20		3.4	4.4.5.1.4
	Deflection Temperature under a load at 1830	20		3.4	4.4.5.1.5

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026/10-23T03:42Z
Check the source to verify that this is the current version before use.

4.4 Quality conformation inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078 and the following as applicable.

4.4.1.1 Obturator lot. A lot shall consist of obturators produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revisions, same specification and same specification revision. Each lot shall contain: bands manufactured from one batch or "run" of resin and molded in one mold on one machine.

4.4.2 Examination. See MIL-A-48078. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:42Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

MIL-O-70548 (AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Ring, Obturating	1 1	9310339
			NEXT HIGHER ASSEMBLY: 9381093/9327838/9310341

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined.				
Major					
101	Ring gage fails to pass freely over the outside diameter of the ring while placed on cylindrical plug		0.40%	3.2	Gage
102	Outside diameter		0.40%	3.2	Gage
103	Width		0.40%	3.2	Gage
104	Thickness		0.40%	3.2	Visual
105	Presence of voids		0.40%	3.3	Visual
106	Color and transparency				
Minor					
201	Surface finish improper		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.6	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026/10-23T03:42Z
Check the source to verify that this is the current version before use.

4.4.3 Testing.

4.4.3.1 Physical properties. Representative specimens shall be randomly selected from each lot of plastic material utilized for fabricating the first article and first two production lots of obturating rings. Five specimen samples shall be subjected to each physical property test listed in 4.4.5.1. If any sample fails to meet any of the requirements specified in section 3, the lot of the plastic material shall be rejected.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.4 herein.

4.4.5 Methods of inspection.

4.4.5.1 Physical characteristics. The specimens shall be subjected to test methods as defined in 4.4.5.1.1 through 4.4.5.1.5 to verify compliance with applicable requirements.

4.4.5.1.1 Izod impact strength. Izod impact strength shall be determined in accordance ASTM-Method D256.

4.4.5.1.2 Tensile yield strength. Tensile yield strength shall be determined in accordance with ASTM-Method D638.

4.4.5.1.3 Flow rate. Flow rate shall be determined in accordance with ASTM-Method D1238.

4.4.5.1.4 Flexural modulus. Flexural modulus shall be determined in accordance with ASTM-Method D790.

4.4.5.1.5 Deflection temperature. Deflection temperature under a load at 1830 kpa shall be determined in accordance with ASTM-Method D648.

5. PACKAGING

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be in accordance with the requirements of Level C of MIL-STD-1169.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:42Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for assembly to M853 illuminating cartridges, M819 Smoke Cartridges and M879 Practice Cartridges for use in the 81mm Mortar.

6.2 Ordering data. See MIL-A-48078 (AR).

6.3 Drawings. Drawings listed in section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by and identified as US Army Armament Research and Development Command (ARRADCOM), Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal Drawings. Technical data originally prepared by the activities is now under cognizance of ARDEC.

6.4 Submission of designs for approval. See para 6.2.3 of MIL-A-48078 (AR). Submit designs as required to Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-I (D), Dover, NJ 07801-5001.

6.5 Subject term (key word) listing.

Deflection temperature
Flexural modulus
Flow rate
Izod impact
Obturators
Tensile strength

Custodian
Army-AR

Preparing activity:
Army-AR

(Project 1395-A226)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:42Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:42Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

COMMANDER
US ARMY ARMAMENT RSCH DEV & ENGR CENTER
ATTN: AMSMC-QA
DOVER, NJ 07801 5001

Source: https://assist.dla.mil -- Downloaded: 202610-23T03:42Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-O-70548	OBTURATING BAND, PLASTIC SPLIT (MOLDED)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:42Z
Check the source to verify that this is the current version before use.